A Unified Framework for Adsorption Isotherm Analysis in Water Quality Studies: Descriptors, Model Selection, and Reporting Guidelines

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Abstract

Adsorption isotherms are central to water-quality research, yet their fitted parameters are often compared across models as if they were physically interchangeable. This practice can be misleading because common isotherm equations differ in their asymptotic behavior, parameter dimensions, and assumptions about surface heterogeneity. Here, we propose a unified framework for adsorption isotherm analysis aimed at making capacity, affinity, and heterogeneity descriptors comparable across adsorbents, contaminants, and studies. The framework separates fitted model parameters from reportable physical descriptors and defines a minimum descriptor set consisting of the saturation capacity Q_{max}, a dimensionless standard-state affinity K^*_{aff}, the corresponding adsorption-energy descriptor E_{ads}, and a reported heterogeneity width σ_H. We show that Q_{max} should be reported only for models with a finite saturation plateau, whereas non-saturating models such as Freundlich cannot provide a physically meaningful maximum capacity. Model-dependent affinity constants are recast through a declared standard-state concentration C^0, allowing thermodynamic comparison on a common scale. Surface heterogeneity is treated through the energy distributions implied by each isotherm family. In addition, we provide a practical workflow combining non-linear regression, residual metrics, information criteria, F-tests, physical decision rules, and optional KNN/SVM-based material classification. Together, these guidelines provide a reproducible route from raw isotherm data to statistically validated, physically interpretable, and cross-study comparable adsorption descriptors for water-treatment materials.

1 Introduction

Access to safe water is becoming one of the defining challenges of this century. Global population reached approximately 8.2 billion in 2024 and is projected to approach 9.7 billion by 2050.[1] Yet, 2.2 billion people still lack safely managed drinking water services.[2] In addition, roughly half of the world's population experiences severe water scarcity for at least part of the year.[3] Urbanization, industrial activity, agricultural intensification, and climate change are compounding these pressures on global water systems.[4] Within this context, the United Nations Sustainable Development Goal 6 (SDG 6), "Clean Water and Sanitation," has catalyzed broad scientific and policy interest in water treatment technologies. However, recent global assessments indicate that the world is not on track to achieve SDG 6 by 2030.[5]

Among the available remediation strategies, adsorption remains especially attractive because of its operational simplicity, compatibility with decentralized treatment, and applicability across a wide range of contaminant classes. Historically, adsorption research in water purification has focused on metals, metalloids, dyes, and pesticides.[6–8] The contaminant landscape, however, is expanding rapidly and now extends far beyond these classical targets. Contemporary challenges increasingly involve persistent and mobile organic pollutants, pharmaceutical residues, endocrine-disrupting compounds, microplastic-associated chemicals, and, most prominently, per- and polyfluoroalkyl substances (PFAS).[9–12] PFAS have become emblematic of the next generation of adsorption targets: they are highly persistent, often mobile in water, resistant to conventional degradation, and subject to tightening regulatory limits in drinking water worldwide.[13–15] As the diversity of target contaminants grows, so does the need for rigorous, cross-pollutant comparison of adsorbent performance, a task that demands a common descriptive language for adsorption data.[16,17]

Adsorption isotherms are the standard tool for summarizing material performance and for extracting descriptors such as maximum uptake capacity, affinity, and site heterogeneity. In practice, however, parameters obtained from different isotherm models are routinely tabulated side by side as though they were interchangeable. They are not. The Langmuir, double Langmuir, Freundlich, and Sips equations arise from different physical assumptions and assign different meanings, dimensions, and asymptotic behaviors to their fitted constants.[18–20] Comparing a Langmuir Q_{max} with a value extrapolated from a Freundlich fit, or placing a Langmuir affinity constant next to a Sips affinity constant whose units depend on the heterogeneity exponent, is neither mathematically consistent nor physically meaningful.[21,22] What often appears to be a comparison of materials is, in reality, a comparison of incompatible parameterizations.[16]

This problem is not merely academic. As adsorption studies move toward application-driven screening of advanced sorbents for complex water matrices—including PFAS, pharmaceuticals, and other emerging contaminants—materials are increasingly ranked as “better” on the basis of larger fitted constants derived from different models, concentration scales, and assumptions about site heterogeneity. Such practice weakens reproducibility, complicates translation from laboratory results to engineering design, and, most importantly, prevents the community from building reliable structure–performance relationships across adsorbents and pollutants. A field that aims to address global water challenges, including those highlighted by SDG 6 and the rapidly evolving contaminant landscape, requires reporting standards that make adsorption parameters comparable in a thermodynamically meaningful way.

In this work, we propose a practical reporting framework grounded in four principles. First, capacity comparisons should be restricted to isotherm models that define a finite saturation capacity. Second, model-dependent affinity constants should be converted, whenever possible, into a dimensionless standard-state affinity and the corresponding adsorption-energy descriptor. Third, model selection should be based on residual diagnostics, information criteria, nested-model tests, and physical admissibility, rather than on R^2 alone. Fourth, once reliable descriptors have been extracted, they can be used for optional downstream classification or clustering of adsorption systems. The mathematical derivations are provided in the SI, and the companion notebooks illustrate the workflow with synthetic examples and user-supplied data.

2 The problem: incompatibility of isotherm-derived parameters

2.1 Common isotherm models and their assumptions

The most widely used isotherm equations in aqueous-phase adsorption can be written as follows. A complete list is included in Table 1 and SI.

Langmuir. Assumes a homogeneous surface with a single class of equivalent sites and monolayer coverage:

$$q_e = \frac{Q_{\max} K_L C_e}{1 + K_L C_e}. \tag{1}$$

Parameters: $Q_{\max}$ ($\mathrm{mg\,g^{-1}}$), the finite saturation capacity, and K_L ($\mathrm{L\,mg^{-1}}$), the affinity constant. See Eq. (S-3).

Double Langmuir. Extends Langmuir to two independent site populations:

$$q_e = \frac{Q_1 K_1 C_e}{1 + K_1 C_e} + \frac{Q_2 K_2 C_e}{1 + K_2 C_e}. \tag{2}$$

Total capacity $Q_{\max} = Q_1 + Q_2$ remains finite and well defined. See Eq. (S-14).

Freundlich. An empirical power law that describes heterogeneous adsorption without a saturation plateau:

$$q_e = K_F C_e^{1/n}. \tag{3}$$

K_F has concentration-dependent units $(\mathrm{mg\,g^{-1}})(\mathrm{L\,mg^{-1}})^{1/n}$; no $Q_{\max}$ is defined. See Eq. (S-22).

Sips (Langmuir–Freundlich). Introduces surface heterogeneity through an exponent m while retaining a saturation plateau:

$$q_e = \frac{Q_{\max} K_S C_e^m}{1 + K_S C_e^m}. \tag{4}$$

When $m = 1$, Sips reduces to Langmuir. The affinity constant K_S carries units that depend on m, i.e. $(\mathrm{L\,mg^{-1}})^m$, and is therefore not directly comparable with K_L. See Eq. (S-29).

> **Parameterization note.** In this manuscript, the Sips constant is defined outside the concentration exponent, $K_S C_e^m$, consistently in the main text, SI, and fitting code. With this convention K_S carries units of $[C]^{-m}$, and the reciprocal-concentration affinity used for descriptor extraction is $K_S^{1/m}$.

As shown in Figures 1–3, the model selection procedure combines physical screening, statistical comparison, and physicochemical validation.

Table 1: Availability of $Q_{\max}$, K_H, and an effective affinity constant K_{aff} for common isotherm models, together with their implied energy-distribution family. The "Recommended use" column is YES when both $Q_{\max}$ and a comparable K_{aff} can be reported, and NO otherwise. See the SI sections in the last column for the derivations and definitions.

Model	$Q_{\max}$	K_H	K_{aff}	**Recommended**	**Energy distribution**	**SI ref.**
Henry	No	Yes	No	NO	– (low-coverage limit)	S-1.1
Langmuir	Yes	Yes	Yes	YES	Dirac δ	S-1.2
Double Langmuir	Yes	Yes	Yes	YES	Two Dirac δ	S-1.3
Freundlich	No	Cond.	No	NO	Power law	S-1.4
Sips (L–F)	Yes	Cond.	Yes *	YES	Logistic	S-1.5
Tóth	Yes	Yes	Yes	YES	Gen. logistic (asym.)	S-1.6
Jovanovic	Yes	Yes	Yes	YES	Gumbel (EV-I)	S-1.7
Temkin	No	No	Yes *	NO	Uniform (rect.)	S-1.8
UNILAN	Yes	Yes *	Yes	YES	Uniform (rect.)	S-1.9
Redlich–Peterson	Cond.	Yes	Yes *	NO	Non-standard (asym.)	S-1.10
Koble–Corrigan	Yes	Cond.	Yes *	YES	Logistic (Sips like)	S-1.11
Radke–Prausnitz	No	Yes	Cond. *	NO	Non-standard (asym.)	S-1.12
D-R / D-A	Yes	No	Yes *	YES	Rayleigh / Weibull	S-1.13
Hill	Yes	Cond.	Yes *	YES	Logistic (Sips like)	S-1.14

Notes. "Cond." indicates availability only in special parameter limits (e.g., $m = 1$, $n_H = 1$, $\beta = 1$). "*." indicates an effective or operational definition; see the SI for explicit formulas and limits.

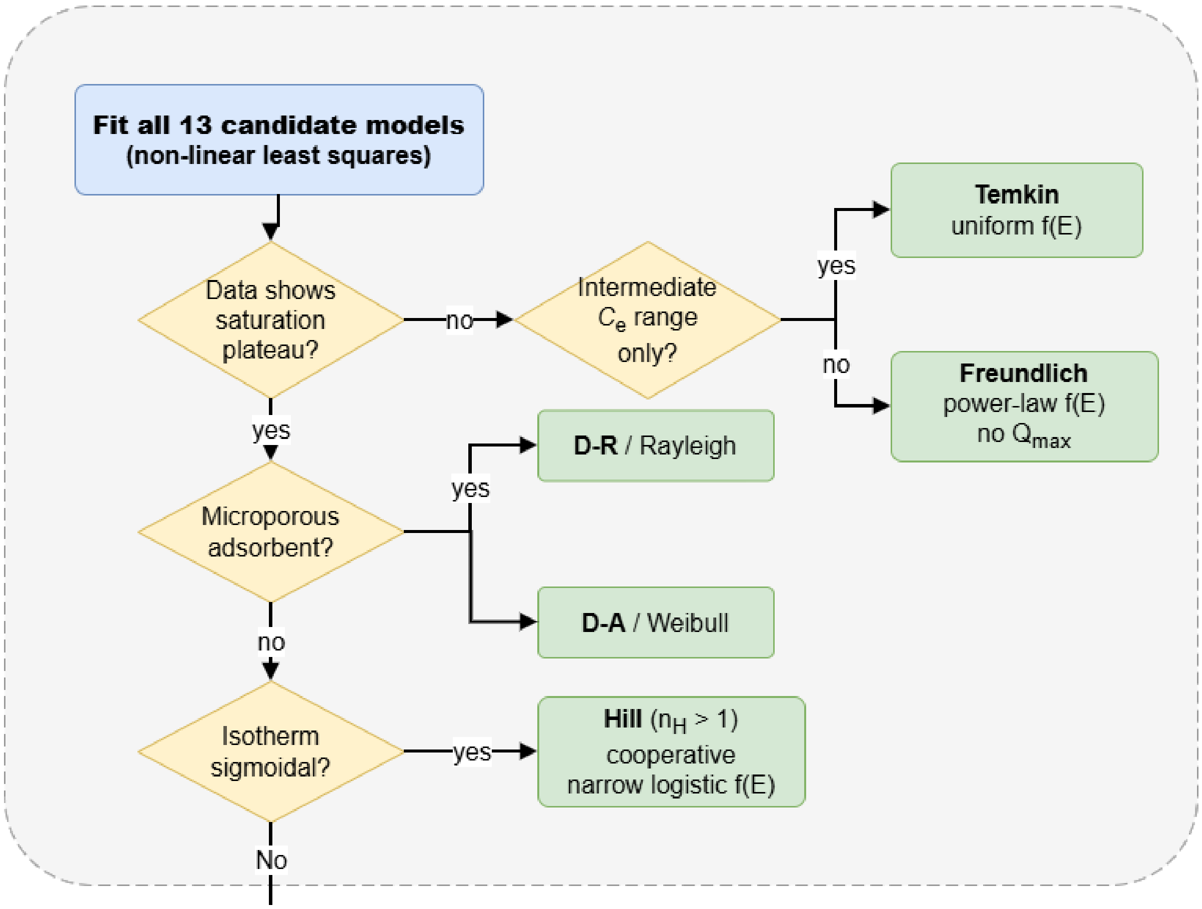

Figure 1: Shape-based pre-screening of adsorption isotherm models. Experimental isotherms are first fitted with the diagnostic model library by non-linear least squares and then classified according to physically observable features, including the presence of a saturation plateau, microporous uptake behavior, sigmoidal adsorption, and restriction to an intermediate concentration range. This first decision layer separates models that can provide finite capacity descriptors from models that are better interpreted as local, empirical, or special-regime descriptions.

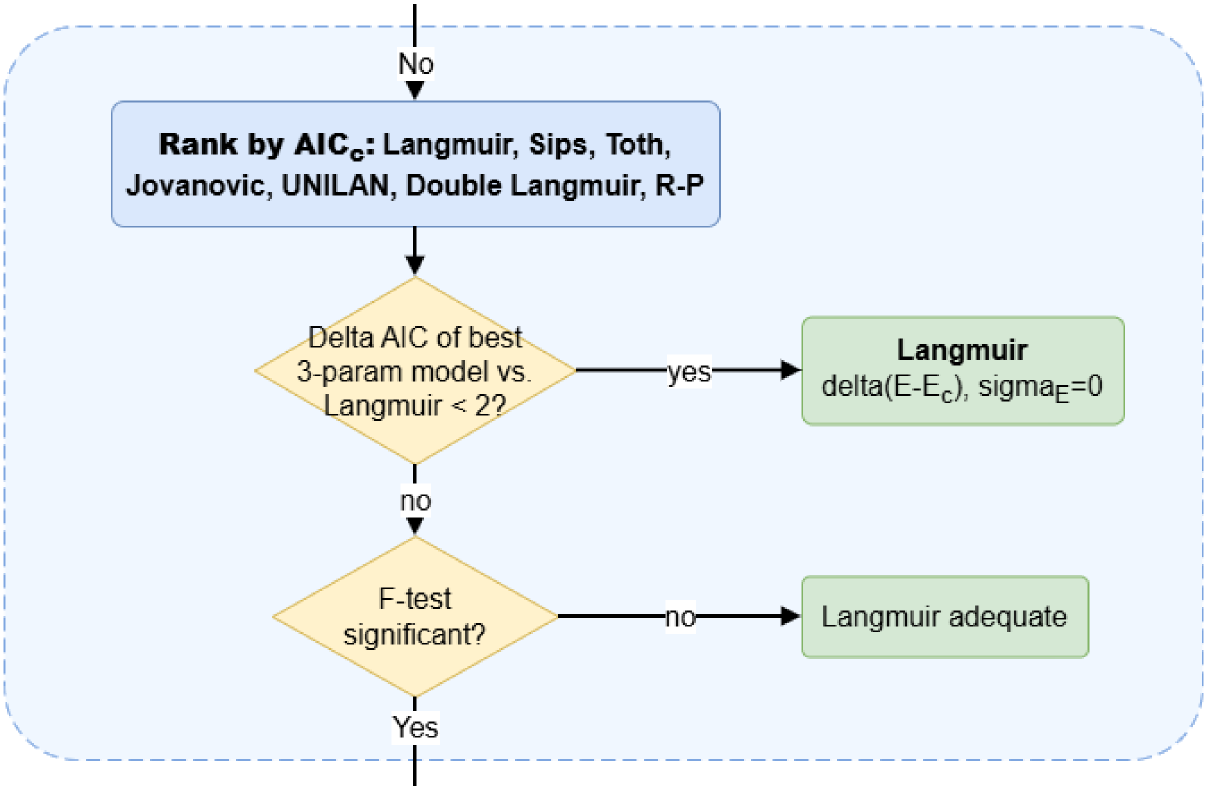

Figure 2: Statistical model-selection step used to decide whether a model more complex than Langmuir is justified. Candidate saturating models are ranked by the corrected Akaike information criterion (AIC_c), and the best three-parameter model is compared against Langmuir. When the improvement in AIC_c is small or the F-test is not significant, the Langmuir model is retained as an adequate description, avoiding over-interpretation of weak heterogeneity.

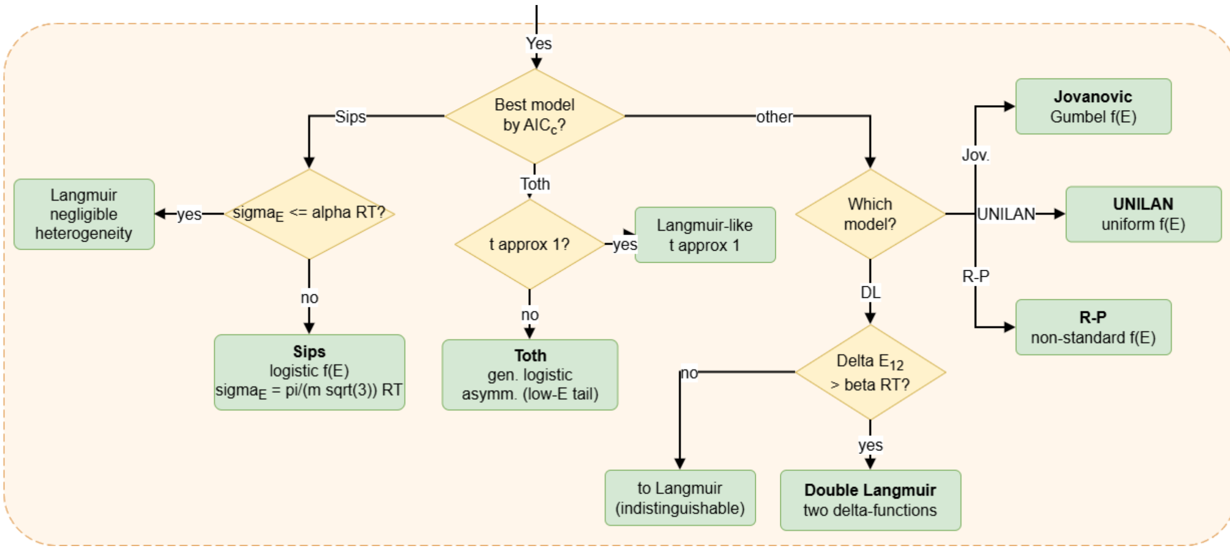

Figure 3: Final physical-validation step for assigning the selected isotherm model to a reportable adsorption descriptor set. The best AIC_c model is interpreted in terms of its implied energy-distribution family and limiting behavior. Sips and Tóth fits are checked for proximity to the Langmuir limit, while Jovanovic, UNILAN, Redlich–Peterson, and double-Langmuir outcomes are assigned according to their characteristic heterogeneity signatures. This step links statistical goodness of fit with physically meaningful descriptors such as finite capacity, affinity, and energy-distribution width.

2.2 Why $Q_{\max}$ values are not always comparable

Even though $Q_{\max}$ is often treated as a universal figure of merit, it is model-specific unless the isotherm explicitly contains a finite saturation plateau. In the Langmuir and Double Langmuir equations, $Q_{\max}$ (or $Q_1 + Q_2$) is a true asymptote: it is the coverage reached as $C_e \to \infty$ and is therefore a well-defined thermodynamic capacity. By contrast, the Freundlich isotherm has no saturation term; q_e increases without bound as C_e increases (SI Eq. (S-22)), so any "$Q_{\max}$" inferred from a Freundlich fit depends entirely on the chosen concentration window. Reporting such a value as a capacity is physically meaningless because it is not a model parameter and has no invariant interpretation.

Even for models that do saturate, comparability depends on whether the saturation regime is actually constrained by the experimental data. In the Sips equation, for example, the heterogeneity exponent m can shift the onset of saturation far beyond the measured concentration range (see Eq. S-29). Under these conditions, the fitted $Q_{\max}$ is not determined by the data but inferred through extrapolation. As a result, reported capacities may reflect model-imposed asymptotes rather than experimentally supported saturation values. The practical consequence is that tables mixing Langmuir or Double Langmuir $Q_{\max}$ values with Freundlich-derived or poorly constrained Sips capacities are not comparing the same physical quantity. Any ranking built on such a mixture conflates experimentally supported saturation capacities with model-dependent extrapolations.

2.3 Why affinity constants are not comparable across models

Affinity constants carry the strongest illusion of comparability because they are reported as single numbers, yet their physical meaning and units change from model to model. In the Langmuir equation, K_L has units of inverse concentration (e.g., $\mathrm{L\,mg^{-1}}$) and sets the midpoint of the uptake curve. In the Sips equation, the constant K_S carries units of $(\mathrm{L\,mg^{-1}})^m$, which depend on the heterogeneity exponent m; therefore its numerical value is not on the same scale as K_L unless $m = 1$ (SI Eqs. (S-3) and (S-29)). The Freundlich constant K_F is even less comparable, because its units depend on $1/n$ and it does not correspond to any saturation behavior (SI Eq. (S-22)).

The Henry constant K_H provides a low-coverage slope (SI Eq. (S-7)), but it is not a full-isotherm descriptor and cannot replace affinity constants derived from different functional forms without additional normalization. As a result, placing K_L, K_S, and K_F side by side implies a common meaning that does not exist. Any comparison based on raw affinity constants is therefore a comparison of incommensurate parameters, not of intrinsic adsorption strength.

3 A common language: standard-state adsorption energy

3.1 Constructing a dimensionless equilibrium constant

To place all isotherm models on a common thermodynamic footing, the model-specific affinity parameter must be expressed as a dimensionless equilibrium constant. This is necessary because the fitted affinity constants used in adsorption isotherms are usually dimensional quantities whose units depend on the mathematical form of the model and on the units chosen for concentration and uptake. Therefore, direct comparison of fitted affinity parameters between different models can be misleading unless a common normalization procedure is applied.

Here, the affinity is defined from the low-concentration Henry region of the isotherm. The Henry constant is obtained as the initial slope,

$$K_H = \lim_{C_e \to 0} \left(\frac{q_e}{C_e} \right), \tag{5}$$

where q_e is the equilibrium uptake and C_e is the equilibrium concentration in solution. For example, if q_e is expressed in $\mathrm{mg\,g^{-1}}$ and C_e in $\mathrm{mg\,L^{-1}}$, then

$$[K_H] = \frac{\mathrm{mg\,g^{-1}}}{\mathrm{mg\,L^{-1}}} = \mathrm{L\,g^{-1}}. \tag{6}$$

Since K_H still contains the adsorbent mass dependence, it is normalized by the maximum adsorption capacity $Q_{\max}$. If $Q_{\max}$ is expressed in $\mathrm{mg\,g^{-1}}$, then

$$\left[\frac{K_H}{Q_{\max}} \right] = \frac{\mathrm{L\,g^{-1}}}{\mathrm{mg\,g^{-1}}} = \mathrm{L\,mg^{-1}}. \tag{7}$$

Thus, $K_H/Q_{\max}$ has units of inverse concentration. A dimensionless affinity constant is then obtained by multiplying this quantity by a declared standard-state concentration C^0

expressed in the same concentration units as C_e:

$$K^*_{\mathrm{aff}} = \frac{K_H}{Q_{\mathrm{max}}}\, C^0. \tag{8}$$

For instance, if C^0 is expressed in $\mathrm{mg\,L^{-1}}$, then

$$[K^*_{\mathrm{aff}}] = \left(\mathrm{L\,mg^{-1}}\right)\left(\mathrm{mg\,L^{-1}}\right) = 1. \tag{9}$$

Therefore, K^*_{aff} is dimensionless by construction. For comparisons among different adsorbates, the declared standard state should preferably be expressed on a molar concentration scale or, when available, in terms of activity. Concentrations in $\mathrm{mg\,L^{-1}}$ are acceptable for water-quality reporting and engineering screening, but they should be interpreted as operational descriptors rather than as composition-independent thermodynamic standard states.For models such as Sips or Koble–Corrigan, where the initial slope at $C_e \to 0$ diverges (i.e., a finite Henry constant does not exist), an operational affinity constant is defined instead by raising the native affinity parameter (carrying fractional units) to the power of the reciprocal exponent (e.g., $K_S^{1/m}$), thereby restoring the units of inverse concentration before normalization.

In this construction, Q_{max} removes the dependence of the Henry slope on the total adsorption capacity, while C^0 removes the remaining concentration unit. Thus, K^*_{aff} represents the standard-state-normalized affinity of the adsorbent–adsorbate pair relative to a defined concentration standard. For the Langmuir isotherm, for example, $K_H = Q_{\mathrm{max}} K_L$, and therefore $K^*_{\mathrm{aff}} = K_L C^0$, recovering the usual dimensionless form of the Langmuir equilibrium constant. The same procedure can be applied to any isotherm model for which a finite Henry limit and a meaningful saturation capacity can be defined.

It should be emphasized that the numerical value of K^*_{aff} depends on the chosen standard state C^0. Therefore, C^0 must be explicitly reported. However, once the same C^0 is used consistently, the resulting dimensionless constants allow affinity parameters extracted from different isotherm models to be compared on the same basis and used in standard thermodynamic expressions.

3.2 Standard-state adsorption energy E_{ads}

The standard Gibbs free energy associated with the declared standard state is

$$\Delta G^0_{\mathrm{ads}} = -RT \ln K^*_{\mathrm{aff}}. \tag{10}$$

For reporting, we use the positive affinity-energy descriptor

$$E_{\mathrm{ads}} \equiv E_{\mathrm{aff}} = -\Delta G^0_{\mathrm{ads}} = RT \ln K^*_{\mathrm{aff}}. \tag{11}$$

Thus, larger E_{ads} corresponds to stronger standard-state affinity. Because K^*_{aff} is dimensionless, the logarithmic term is dimensionless and the units of E_{ads} arise only from RT:

$$[E_{\mathrm{ads}}] = [RT] = \mathrm{kJ\,mol^{-1}} \tag{12}$$

when R is expressed in $\mathrm{kJ\,mol^{-1}\,K^{-1}}$. The descriptor is therefore a standardized model-derived measure of affinity, not a raw fitted parameter and not a directly measured observable independent of the fitted model.

3.3 Extraction of K^*_{aff} and E_{ads} from each model

Table 2 shows how K^*_{aff} and E_{ads} are obtained from the native parameters of each isotherm.

Table 2: Extraction of native and operational affinity descriptors from isotherm parameters. C^0 is the declared standard-state concentration; descriptors are reportable only when the fitted model is physically admissible and identifiable.

Model	$\boldsymbol{Q_{\max}}$	$\boldsymbol{K_{\mathrm{aff}}}$ **(native)**	$\boldsymbol{E_{\mathrm{ads}}/(RT)}$
Langmuir	$Q_{\max}$	K_L	$\ln(K_L\, C^0)$
Double Langmuir	$Q_1 + Q_2$	$K_L^{(\mathrm{tot})} = \dfrac{Q_1 K_1 + Q_2 K_2}{Q_1 + Q_2}$	$\ln(K_L^{(\mathrm{tot})}\, C^0)$
Sips	$Q_{\max}$	$K_S^{1/m}$	$\frac{1}{m} \ln(K_S\, (C^0)^m)$
Tóth	$q_{\max}$	b	$\ln(b\, C^0)$
Jovanovic	$q_{\max}$	b	$\ln(b\, C^0)$
Temkin	—	a_T	$\ln(a_T\, C^0)$
UNILAN	$Q_{\max}$	$\sqrt{b_{\max} b_{\min}}$	$\frac{1}{2} \ln(b_{\max} b_{\min}\, (C^0)^2)$
Redlich–Peterson	K_{RP}/a_{RP} [a]	a_{RP} [a]	$\ln(a_{RP}\, C^0)$
Koble–Corrigan	A_{KC}/B_{KC}	B_{KC}^n	$n \ln(B_{KC}\, C^0)$
Radke–Prausnitz	—	r_{RP}	$\ln(r_{RP}\, C^0)$
D-R / D-A	$Q_{\max}$	$1/C_{1/2}$ [b]	$\ln(C^0/C_{1/2})$
Hill	$Q_{\max}$	K_D^{-1/n_H}	$-\frac{1}{n_H} \ln K_D + \ln C^0$

[a] Only when $\beta = 1$. [b] $C_{1/2}$ is the half-loading concentration; see Sec. S-1.13. "—" indicates no finite $Q_{\max}$ exists.

The Freundlich isotherm may describe adsorption data over a restricted concentration window, but it is absent from Table 2 because it defines no finite $Q_{\max}$. Consequently, the ratio $K_H/Q_{\max}$ is undefined and no universal dimensionless affinity can be constructed from the Freundlich parameters alone. Only a local, reference-dependent apparent affinity can be defined after choosing a concentration C_{ref}, and this quantity should not be reported as a universal descriptor. For the Double Langmuir model, an effective total adsorption energy can be reported from the capacity-weighted constant $K_L^{(\mathrm{tot})}$, while the site-resolved constants K_1 and K_2 remain necessary to preserve the bimodal interpretation captured by the site-resolved energy separation. Finally, the choice of standard-state concentration C^0 must be explicitly declared; although it appears inside the logarithm and therefore cancels in any *difference* ΔE_{ads}, it must be consistent across all entries in a comparative table. The full set of energy-distribution families and their relation to σ_E is given in Table S-1; numerical values of σ_E at $T = 298$ K are collected in Table S-2.

4 Quantifying surface heterogeneity: energy distributions

4.1 From adsorption sites to observable energy distributions

The final variable in the adsorption system is the distribution of site energies. Unlike Q_{max} or E_{ads}, this quantity is usually not fitted explicitly, yet it is implicitly encoded in every isotherm model. In other words, each adsorption equation corresponds to a particular assumption about how adsorption energies are distributed across the surface. This provides a natural way to place different isotherm models in a common physical framework.

Real adsorbent surfaces are energetically heterogeneous because adsorption sites differ in local chemistry, accessible geometry, coordination environment, and confinement. As a result, adsorption is more realistically described by a distribution of site energies than by a single characteristic value. A heterogeneous surface can therefore be represented as an ensemble of independent Langmuir-type sites, each characterized by an adsorption energy E, so that the macroscopic isotherm is written as

$$q_e = Q_{\mathrm{max}} \int_{-\infty}^{\infty} \Theta_L(E - E_c)\, f(E)\, \mathrm{d}E, \tag{13}$$

where $f(E)$ is the normalized distribution of adsorption site energies,

$$\int_{-\infty}^{\infty} f(E)\, \mathrm{d}E = 1, \tag{14}$$

and $\Theta_L(E - E_c)$ is the Langmuir occupation kernel,

$$\Theta_L(E - E_c) = \frac{1}{1 + \exp[-(E - E_c)/(RT)]}. \tag{15}$$

Here E_c is the condensation energy associated with the equilibrium concentration,

$$E_c = RT \ln\left(\frac{1}{K_0 C_e}\right), \tag{16}$$

with K_0 a reference constant used to define the energy scale.

Equation (13) emphasizes an important point: the experimentally observed isotherm does not probe the intrinsic distribution $f(E)$ directly, but rather the convolution of that distribution with the thermal occupation kernel. Differentiating eq (13) with respect to E_c gives the observable energy-density profile

$$\phi(E_c) = -\frac{1}{Q_{\max}} \frac{\mathrm{d}q_e}{\mathrm{d}E_c} = \int_{-\infty}^{\infty} f(E)\, g_T(E - E_c)\, \mathrm{d}E, \tag{17}$$

where

$$g_T(x) = \frac{\exp(-x/RT)}{RT\,[1 + \exp(-x/RT)]^2} \tag{18}$$

is the thermal broadening kernel. This kernel is a logistic probability density with standard deviation

$$\sigma_T = \frac{\pi}{\sqrt{3}} RT. \tag{19}$$

Thus, even a perfectly homogeneous Langmuir surface, intrinsically described by a Dirac distribution,

$$f(E) = \delta(E - E_1), \tag{20}$$

is observed through the broadened profile

$$\phi(E_c) = g_T(E_1 - E_c), \tag{21}$$

which already has a finite width set by temperature. At $T = 298$ K, this intrinsic thermal width is

$$\sigma_T \approx 4.5 \text{ kJ mol}^{-1}. \tag{22}$$

This provides a physical interpretation for why systems with very small energetic dispersion often appear experimentally indistinguishable from Langmuir behavior.

The commonly used condensation approximation replaces the smooth Langmuir kernel by a step function,

$$\Theta_L(E - E_c) \approx \Theta(E - E_c), \tag{23}$$

so that thermal broadening is neglected and the intrinsic distribution can be recovered directly as

$$f(E) = -\frac{1}{Q_{\max}} \frac{\mathrm{d}q}{\mathrm{d}E}. \tag{24}$$

This approximation is useful because it provides a practical route from any empirical isotherm equation back to its implied energy distribution. Full derivations for the models treated here are given in Section S-4.4.

4.2 Discrete sites, thermal broadening, and continuous heterogeneity

The relation between intrinsic and observable distributions clarifies how the usual isotherm families are connected. Consider first a surface with two discrete classes of sites,

$$f(E) = w\,\delta(E - E_1) + (1 - w)\,\delta(E - E_2), \qquad 0 \le w \le 1. \tag{25}$$

Under thermal broadening, the experimentally accessible profile becomes

$$\phi(E_c) = w\,g_T(E_1 - E_c) + (1 - w)\,g_T(E_2 - E_c). \tag{26}$$

If the two site energies are sufficiently separated relative to the thermal broadening scale, the bimodal character is preserved and a Double Langmuir description remains physically meaningful. For the symmetric case of equal populations, the onset of bimodality for the logistic kernel occurs at

$$|E_1 - E_2| > 2RT\,\mathrm{arcosh}(2) \approx 2.63\,RT \approx 1.45\,\sigma_T. \tag{27}$$

Below this scale, the two thermally broadened contributions overlap strongly and may appear as a single broadened population. In that regime, the same data may be described by a continuous heterogeneous model such as Sips or Tóth rather than by two explicitly resolved Langmuir populations. For unequal populations, the exact threshold also depends on the statistical weights of the two site classes.

This interpretation also explains the role of temperature. Increasing T increases the kernel width σ_T according to eq (19), making nearby site populations less distinguishable. Conversely, lower temperatures sharpen the energetic resolution and make multimodal structure easier to detect. Therefore, the observed heterogeneity is not only a property of the surface itself, but also of the thermal scale at which the experiment is performed.

In this sense, the standard isotherm families can be viewed as limiting cases of the same general picture. Langmuir corresponds to a single discrete energy level. Double Langmuir corresponds to two discrete levels. Sips, Hill, and Koble–Corrigan correspond to symmetric continuous distributions with finite width. Tóth and Jovanovic correspond to asymmetric continuous distributions. Freundlich represents the limiting case of extremely broad or effectively unbounded heterogeneity. The different models are therefore not disconnected empirical formulas, but alternative representations of how adsorption energies are distributed and thermally resolved.

4.3 Energy-distribution families implied by common isotherms

Applying eq (24) to each isotherm yields the energy-distribution families summarized in Table 3. These distributions should be interpreted as the model-implied site-energy distributions under the condensation approximation.

Table 3: Energy-distribution families and apparent mathematical width σ_E implied by each isotherm model. Parameter m (Sips, Koble–Corrigan) is the heterogeneity exponent; t (Tóth) and n_H (Hill) are the respective shape parameters. σ_E is in units of RT ($RT \approx 2.48\,\mathrm{kJ\,mol^{-1}}$ at 298 K). Full derivations are given in Section S-4.4.

Model	**Distribution $f(E)$**	**Symm.**	$\boldsymbol{\sigma_E/(RT)}$
Langmuir	Dirac $\delta(E - E_c)$	—	0
Double Langmuir	Two Dirac δ	—	$\frac{\sqrt{Q_1 Q_2}}{Q_1+Q_2}\frac{\lvert\Delta E_{12}\rvert}{RT}$
Sips	Logistic (scale RT/m)	Yes	$\pi/(m\sqrt{3})$
Tóth	Generalized logistic type IV	No	$> \pi/\sqrt{3}$ ($t < 1$); no closed form
Jovanovic	Gumbel (EV-I)	No	$\pi/\sqrt{6} \approx 1.28$
Temkin	Uniform	Yes	$b_T Q_{\max}/(2\sqrt{3}\,RT)$
UNILAN	Uniform (exact kernel)	Yes	$\ln(b_{\max}/b_{\min})/(2\sqrt{3})$
Freundlich	Power law (unbounded)	—	∞
Redlich–Peterson	Non-standard ($\beta < 1$)	No	—
Koble–Corrigan	$\equiv$ Sips	Yes	$\pi/(m\sqrt{3})$
Radke–Prausnitz	Non-standard	No	—
D-R ($n_{DA} = 2$)	Rayleigh	No	$0.463\,E_a/(RT)$
D-A (general)	Weibull (shape n_{DA})	No	$(E_a/RT)\sqrt{\Gamma(1 + 2/n_{DA}) - [\Gamma(1 + 1/n_{DA})]^2}$
Hill	$\equiv$ Sips (scale RT/n_H)	Yes	$\pi/(n_H\sqrt{3})$

Several physically meaningful relations follow from Table 3. First, Langmuir and Double Langmuir represent discrete energetic populations, whereas Sips, Hill, Koble–Corrigan, Tóth, Jovanovic, Temkin, and UNILAN represent continuous distributions. Second, Sips, Hill, and Koble–Corrigan belong to the same symmetric logistic family and differ only in parameterization. Third, Tóth and Jovanovic introduce skewness, showing that heterogeneity is not fully described by width alone. Fourth, Freundlich corresponds to a limiting case in which the energy dispersion becomes effectively unbounded, consistent with the absence of a finite saturation capacity. Finally, the Dubinin family is naturally expressed in the Polanyi adsorption potential rather than directly in E, but still admits an equivalent probabilistic interpretation through Weibull-type distributions.

4.4 Physical and statistical interpretation of the energy distribution

Once an isotherm has been mapped onto an energy distribution, two widths must be kept conceptually separate. The quantity σ_E denotes the apparent mathematical width of the model-implied energy distribution. The reported heterogeneity descriptor is σ_H, which removes the homogeneous thermal contribution whenever that correction is defined. Therefore, Langmuir is recovered at $\sigma_H = 0$, even though thermally broadened occupation kernels may have a finite apparent width. Representative numerical values and the model-specific corrections are given in Table S-2.

This representation also provides a natural connection to statistical model comparison. Two different isotherm models may fit the same data well because their implied energy distributions have similar central energies and similar widths, even if their algebraic forms differ. For example, a narrow bimodal distribution and a single broader logistic distribution may generate nearly indistinguishable isotherms when the two discrete energies are closer than the thermal resolution. Conversely, two models with similar residual error but markedly different E_{ads} or σ_E imply different physical pictures of the surface.

The similarity between two implied energy distributions, $f_1(E)$ and $f_2(E)$, can be quantified directly through an overlap coefficient,

$$\text{OVL} = \int_{-\infty}^{\infty} \min[f_1(E), f_2(E)]\, \mathrm{d}E, \qquad 0 \leq \text{OVL} \leq 1. \tag{28}$$

High overlap indicates that two models encode essentially the same energetic picture, whereas low overlap indicates physically distinct heterogeneity patterns.

More broadly, the descriptor set

$$\mathcal{F} = \{Q_{\max},\, K^*_{\text{aff}},\, E_{\text{ads}},\, \sigma_H\}\,. \tag{29}$$

defines a compact, physically interpretable feature space for adsorption systems. Within this space, $Q_{\max}$ measures the amount of adsorption, E_{ads} measures the central energetic level, and σ_H measures the reported model-implied heterogeneity. These descriptors can be used not only to compare fitted models, but also to compare materials, contaminants, or operating conditions on a common basis. They also provide suitable input variables for data-driven analysis, including clustering and supervised classification methods such as k-nearest neighbours (KNN) and support vector machines (SVM), as developed in the following section. In this way, the energy-distribution view links isotherm physics, statistical model discrimination, and descriptor-based classification within a single framework.

5 Model selection and statistical comparison

5.1 Non-linear regression

To extract the proposed standardized reporting descriptors, the diagnostic model library should be fitted to the experimental q_e–C_e data by non-linear or weighted non-linear regression (e.g. Levenberg–Marquardt or trust-region algorithms), minimising the sum of squared residuals $\text{SSE} = \sum_i (q_i^{\text{exp}} - q_i^{\text{pred}})^2$ when errors are homoscedastic. When experimental uncertainties or non-constant variance are available, weighted non-linear regression should be used with the corresponding likelihood. Linearised forms (e.g. $1/q_e$ vs. $1/C_e$ for Langmuir, $\ln q_e$ vs. $\ln C_e$ for Freundlich) must be avoided because they distort the error structure and introduce bias in the estimated parameters. Physical bounds should be imposed on all parameters ($Q_{\max} > 0$, $K > 0$, $0 < m \leq 1$ for Sips, etc.), and multiple initial guesses should be tested to guard against local minima. Numerical convergence alone is not considered model acceptance. The full model library is used as a diagnostic screen; final interpretation is restricted to models that are physically admissible, identifiable from the available data, and statistically plausible.

Table 4: Model status used in the diagnostic fitting workflow.

Status	Meaning	Used for reporting?
Fitted	Numerical convergence was achieved	No
Admissible	Parameters and limits are physically meaningful	Maybe
Identifiable	Parameters are constrained by the data and bootstrap-stable	Maybe
Plausible	Within statistical support, e.g. $\Delta\mathrm{AIC}_c \leq 2$	Yes, as candidate
Reported	Simplest physically meaningful model or descriptor range	Yes

5.1.1 Residual error metrics

No single error metric captures all aspects of fit quality; therefore, multiple complementary metrics should be reported. The complete equations are given in Section S-2.1; here we summarise their purpose and interpretation.

- **SSE** (sum of squared errors, Eq. S-106): the objective function minimised during fitting. Sensitive to large deviations (outliers).
- R^2 (coefficient of determination, Eq. S-107): fraction of variance explained. Values close to 1 indicate a good fit, but R^2 alone cannot distinguish overfitting.
- R^2_{adj} (adjusted R^2, Eq. S-108): penalises model complexity by accounting for the number of parameters p. Preferred over R^2 when comparing models with different numbers of parameters.
- **EABS** (sum of absolute errors, Eq. S-109): less sensitive to outliers than SSE. Useful as a complementary absolute-scale measure.
- **RESID** (mean absolute residual, Eq. S-110): average deviation per data point in the same units as q_e. Provides an intuitive measure of typical prediction error.
- **ARED** (average relative error in percent, Eq. S-111): normalises each residual by q_i^{exp}, giving equal weight to low- and high-concentration points. Particularly informative for data spanning several orders of magnitude.

- **MPSED** (median percentage squared error, Eq. S-112): a robust (median-based) relative error metric, resistant to individual outliers.
- **HYBRID** (Eq. S-113): a compromise between SSE and relative error that penalises underfitting at low concentrations, where q_i^{exp} is small and squared residuals are amplified by the $1/q_i^{\mathrm{exp}}$ weight. It also corrects for degrees of freedom $(n - p)$.

We recommend reporting at least SSE, R^2_{adj}, ARED, and HYBRID for every model. The combination provides sensitivity to both absolute and relative deviations, robustness to outliers, and a degrees-of-freedom correction.

5.2 Information criteria: AIC and BIC

In adsorption studies, model selection is often based on goodness-of-fit metrics such as the coefficient of determination (R^2) or the sum of squared errors (SSE). However, these metrics systematically favor models with a larger number of adjustable parameters, even when the improvement in fit is not physically meaningful. This is particularly problematic when comparing isotherm models of different complexity, such as Langmuir, Sips, or multi-site formulations, where additional parameters can artificially improve the apparent agreement with experimental data.

To address this issue, information criteria provide a principled framework for balancing model fidelity and complexity. These criteria penalize overparameterized models, allowing comparison across non-nested models with different numbers of fitted parameters.

The Akaike Information Criterion (AIC), its small-sample correction (AIC_c), and the

Bayesian Information Criterion (BIC) are defined as:

$$\mathrm{AIC} = n \ln\left(\frac{\mathrm{SSE}}{n}\right) + 2p, \tag{30}$$

$$\mathrm{AIC}_c = \mathrm{AIC} + \frac{2p(p+1)}{n-p-1}, \tag{31}$$

$$\mathrm{BIC} = n \ln\left(\frac{\mathrm{SSE}}{n}\right) + p \ln n, \tag{32}$$

where n is the number of data points and p the number of fitted parameters.

The first term in each expression reflects the goodness of fit through the residual variance, while the second term introduces a penalty that increases with model complexity. AIC applies a constant penalty per parameter, whereas BIC imposes a stronger penalty that scales with $\ln n$, favoring simpler models as the dataset grows.

For small datasets, which are common in adsorption experiments, the corrected form AIC_c is preferred, as it compensates for finite-sample bias and prevents the selection of overly flexible models. AIC_c is meaningful only when $n > p + 1$; models violating this condition are rejected as underdetermined. For weighted fits, the corresponding weighted log-likelihood should be used instead of the unweighted SSE expression.

Models are ranked according to their relative information loss, quantified by the difference

$$\Delta\mathrm{AIC}_c = \mathrm{AIC}_{c,i} - \mathrm{AIC}_{c,\min}. \tag{33}$$

The conventional interpretation is that $\Delta\mathrm{AIC}_c < 2$ indicates substantial support for multiple models, $4 < \Delta\mathrm{AIC}_c < 7$ indicates considerably less support for the higher-AIC model, and $\Delta\mathrm{AIC}_c > 10$ indicates essentially no support.

When several models fall within $\Delta\mathrm{AIC}_c < 2$, model discrimination becomes ambiguous. In such cases, Akaike weights (Eq. S-132) can be used to estimate the relative likelihood of each model given the data. These weights provide a probabilistic interpretation of model support and can be used to perform model-averaged predictions, thereby reducing sensitivity to model selection.

Importantly, the use of information criteria does not imply that one model is the "true" model, but rather identifies the model that achieves the best trade-off between accuracy and parsimony within the candidate set. In the context of adsorption isotherms, this is essential to avoid overinterpreting parameters obtained from overly flexible models that fit noise rather than underlying adsorption behavior.

5.3 F-test for nested models

When a simpler model is nested within a more complex one (e.g. Langmuir $\subset$ Sips, Langmuir $\subset$ Double Langmuir), the F-test determines whether the additional parameters provide a statistically significant improvement:

$$F = \frac{(\mathrm{SSE}_1 - \mathrm{SSE}_2)/(p_2 - p_1)}{\mathrm{SSE}_2/(n - p_2)}. \tag{34}$$

The F-test is treated as an auxiliary criterion because nested comparisons can be delicate when the additional parameter lies near a boundary, such as the Sips limit $m = 1$. Bootstrap identifiability and physical admissibility should confirm any F-test-based decision. Beyond the F-test, physically motivated criteria can guide the choice between nested alternatives:

- **Langmuir vs. Sips:** Accept Langmuir if the Sips-derived intrinsic heterogeneity is negligible, $\sigma_H^{(\mathrm{Sips})} = (\pi/\sqrt{3})RT\sqrt{1/m^2 - 1} \le \alpha RT$, or if m is statistically indistinguishable from 1 within uncertainty (Eq. S-139).

- **Langmuir vs. Double Langmuir:** Accept Double Langmuir if the two site classes are resolvable, $\Delta E_{12} = RT|\ln(K_1/K_2)| > \beta\, RT$ with $\beta \approx 2$–3 (Eq. S-140).

- **Sips vs. Double Langmuir:** Prefer Sips when the site-energy separation is smaller than the distribution width, $\Delta E_{12} < 2\,\sigma_H^{(\mathrm{Sips})}$ (Eq. S-141).

The overlap coefficient (OVL) (Eq. S-136) provides a complementary, distribution-level similarity metric. A complete decision tree integrating all these criteria is given in Figure S-1.

5.4 Downstream descriptor use: k-Nearest Neighbours classification

The KNN and SVM examples are intended as downstream demonstrations of how the standardized descriptor table can be used once a sufficiently large labelled dataset exists. They are not used here to claim predictive performance of a trained classifier.

For classification and clustering, the reporting fingerprint is converted into the standardized feature vector

$$\mathbf{x} = \left(Q_{\max},\, \ln K^{*}_{\text{aff}},\, \sigma_H\right). \tag{35}$$

Here, $Q_{\max}$ captures capacity, $\ln K^{*}_{\text{aff}}$ captures standard-state affinity on an additive scale, and σ_H captures the reported model-implied heterogeneity.

Once the feature vector in Eq. (35) has been extracted for a collection of adsorbent–adsorbate systems, materials can be classified in this descriptor space without imposing a parametric model on the group boundaries. The k-Nearest Neighbours (KNN) algorithm assigns a query material to the class most common among its k closest neighbours:

$$\hat{y}(\mathbf{x}_q) = \arg\max_{c} \sum_{i \in \mathcal{N}_k(\mathbf{x}_q)} \mathbf{1}[y_i = c], \tag{36}$$

where $\mathcal{N}_k(\mathbf{x}_q)$ denotes the set of k nearest neighbours of the query point $\mathbf{x}_q$ in the standardised descriptor space, and $\mathbf{1}[\cdot]$ is the indicator function. Because KNN is non-parametric and requires no assumption about group geometry, it is well suited for exploratory classification tasks such as:

- **Performance grouping:** sorting adsorbents into high-, medium-, and low-capacity tiers based on $(Q_{\max},\, \ln K^{*}_{\text{aff}})$.

- **Material family identification:** given a reference library, predicting whether a new material behaves like an activated carbon, zeolite, MOF, or biochar.

- **Pollutant selectivity:** classifying adsorbent–adsorbate pairs by target contaminant class (e.g. heavy metals, dyes, pharmaceuticals, PFAS).

- **Homogeneity screening:** separating materials with near-homogeneous surfaces ($\sigma_H \approx 0$) from those with broad energy distributions ($\sigma_H \gg RT$).

Figure 4 illustrates the first of these tasks for three synthetic performance groups projected onto the ($Q_{\max}$, $\ln K^*_{\text{aff}}$) plane.

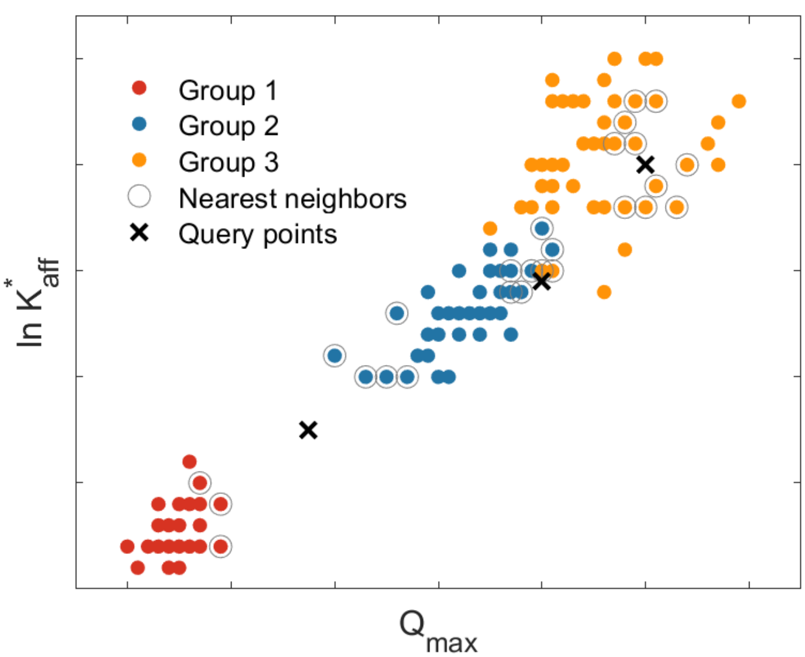

Figure 4: k-Nearest Neighbours classification of adsorbent materials in the standardized descriptor space ($Q_{\max}$, $\ln K^*_{\text{aff}}$). Three performance groups are shown: Group 1 (low capacity), Group 2 (intermediate), and Group 3 (high capacity). Query points ($\times$) are classified by majority vote among their $k = 10$ nearest neighbours (grey circles).**Note**: These synthetic data are shown only to illustrate the classification workflow and should not be interpreted as classifier validation

5.5 Material classification by Support Vector Machines

When the classification task requires a well-defined decision boundary, a Support Vector Machine (SVM) finds the maximum-margin separating hyperplane in descriptor space:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 + C_{\text{reg}} \sum_i \xi_i, \qquad y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \xi_i, \tag{37}$$

where $\mathbf{w}$ is the normal to the hyperplane, b is the bias, ξ_i are slack variables for misclassified points, and C_{reg} controls the trade-off between margin width and classification error. The support vectors—the data points that lie on or within the margin—define the boundary entirely. Figure 5a shows a linear SVM separating two classes of adsorbents in the $(Q_{\text{max}}, \ln K^*_{\text{aff}})$ plane, corresponding to a two-dimensional projection of the feature vector in Eq. (35).

When the full harmonized descriptor set $\{Q_{\text{max}}, \ln K^*_{\text{aff}}, \sigma_H\}$ is used, the SVM operates in three dimensions and the decision boundary becomes a *plane* (Figure 5b). For non-linearly separable data an RBF kernel $\mathcal{K}(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2)$ replaces the linear kernel, producing curved decision surfaces that can capture complex group boundaries while still being defined entirely by the support vectors.

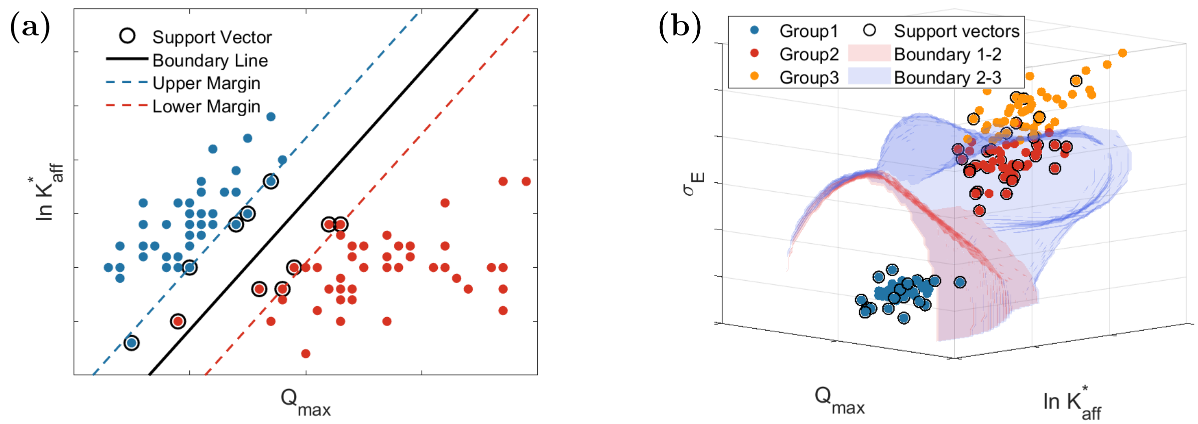

Figure 5: Support Vector Machine classification of adsorbent materials. (a) Two-dimensional projection onto the ($Q_{\max}$, $\ln K^*_{\text{aff}}$) plane: the solid line is the decision boundary $\mathbf{w}\cdot\mathbf{x} + b = 0$; dashed lines indicate the margin boundaries; support vectors are circled. (b) Three-dimensional descriptor space ($Q_{\max}$, $\ln K^*_{\text{aff}}$, σ_H): decision boundaries are now planes (or curved surfaces when an RBF kernel is used). Three performance groups and two inter-group boundaries are shown; support vectors are marked with black circles. **Note**: These synthetic data are shown only to illustrate the classification workflow and should not be interpreted as classifier validation

Categorical variables—such as material type (e.g. activated carbon, zeolite, MOF), best-fit isotherm model, or synthesis route—can be incorporated into the SVM framework through one-hot encoding. Each category is expanded into binary indicator features that enter $\mathbf{x}_i$ alongside the continuous descriptors ($Q_{\max}$, $\ln K^*_{\text{aff}}$, σ_H). In the resulting augmented feature space the SVM decision boundary is no longer a plane but a *hyperplane* whose dimensionality equals the total number of features minus one. For a problem with p continuous descriptors and q one-hot columns the separating surface lives in $\mathbb{R}^{p+q}$ and its projection onto any two-dimensional descriptor pair produces the familiar boundary visualised in Figure 5a. Typical SVM classification tasks in adsorption studies include:

- **Binary screening:** suitable vs. unsuitable adsorbents for a given application (e.g. materials exceeding a regulatory removal threshold for drinking water).
- **Model-type separation:** distinguishing systems that are best described by a Langmuir-type isotherm ($\sigma_H \approx 0$) from those requiring a heterogeneous model (Sips, Tóth).

- **Material-class boundaries:** separating material families (activated carbon vs. zeolite vs. MOF) in the descriptor space, using one-hot encoded material type as additional features.

- **Multi-class ranking:** extending the binary SVM via one-vs-one or one-vs-all strategies to classify adsorbents into three or more performance groups (as in Figure 5b).

Both KNN and SVM operate on the same standardized descriptor table (Eq. S-191) with Euclidean distance (Eq. S-193). Ward's hierarchical clustering (Eq. S-199) provides an unsupervised alternative when class labels are unknown. The key advantage is that all three methods operate on model-harmonized descriptors, ensuring that the grouping reflects standardized model-derived adsorption behavior rather than artefacts of different fitting models. The full mathematical details of all clustering and classification methods are given in Section S-5.

6 Recommended reporting guidelines

The following guidelines distill the preceding analysis into a practical reporting framework. Their goal is not to prescribe a single isotherm model but to ensure that, regardless of which model is selected, the published results are sufficient for cross-study comparison.

6.1 Minimum reporting set

For every adsorbent–adsorbate system analysed, the following quantities should be reported:

1. **Experimental conditions:** temperature T, pH, ionic strength, and the declared standard-state concentration C^0 (with units).

2. **Model fit:** best-fit model name, all fitted parameters $\boldsymbol{\theta}$ with units and confidence intervals, and at least SSE, R^2, ARED, and AIC$_c$.

3. **Reported descriptors:** $Q_{\max}$ ($\mathrm{mg\,g^{-1}}$), K^*_{aff} (dimensionless), E_{ads} ($\mathrm{kJ\,mol^{-1}}$), and σ_H ($\mathrm{kJ\,mol^{-1}}$).

4. **Comparison metrics** (when applicable): OVL between competing model distributions and/or cluster labels from KNN/SVM classification.

The complete minimum reporting table is given in Table S-3; the extraction formulas for each model are collected in Table S-4.

6.2 Recommended analysis workflow

We recommend a seven-step pipeline:

1. **Data collection.** Record q_e–C_e pairs at constant T and pH, with at least $n \geq 8$ points as an absolute minimum, and preferably $n \geq p + 5$ for every candidate model considered, spanning the linear, transition, and plateau regions whenever saturation descriptors are to be reported.

2. **Non-linear fitting.** Fit all models as a diagnostic library, but interpret only the models that are physically admissible, statistically supported, and identifiable from the available data. (Section 5.1).

3. **Error metrics.** Compute SSE, R^2, R^2_{adj}, ARED, and HYBRID for each model (Section S-2.1).

4. **Model selection.** Rank by AIC$_c$; apply F-tests for nested pairs; apply physical decision criteria (Section 5.3, Figure S-1).

5. **Descriptor extraction.** From the selected model, compute $Q_{\max}$, K^*_{aff}, E_{ads}, and σ_H (Table 2).

6. **Statistical comparison.** Compute OVL between energy distributions of alternative models or between different materials (Section S-3.2).

7. **Classification.** Map materials into the standardised descriptor space and apply clustering or supervised classification (Section 5.4, Section 5.5).

6.3 Practical recommendations

- **Never linearise.** Linearised isotherm forms distort the error structure and bias the fitted parameters. Always use non-linear regression.

- **Do not report $Q_{\max}$ from the Freundlich equation.** The Freundlich isotherm has no saturation; any "capacity" derived from it is a concentration-window artefact.

- **Always declare C^0.** The numerical value of E_{ads} depends on C^0; omitting it renders the result non-reproducible. Differences ΔE_{ads} between materials measured under the same C^0 are invariant.

- **Report confidence intervals.** Bootstrap or Monte Carlo uncertainties on $Q_{\max}$, E_{ads}, and σ_H should accompany every table entry.

- **Use σ_H to assess model necessity.** If the best 3-parameter model yields only weak heterogeneity, e.g. $\sigma_H \leq 2RT$ or, for Sips, m remains close to 1 within uncertainty, the Langmuir model is likely sufficient and the extra parameter is not justified (Eq. S-139).

- **Prefer the simplest adequate model.** Among models with $\Delta\text{AIC}_c < 2$, report the one with the fewest parameters.

7 Conclusions

Adsorption isotherms remain indispensable tools for evaluating materials for water-quality applications, but their fitted parameters cannot be treated as universally interchangeable quantities. This work shows that many common comparisons in the adsorption literature mix parameters with different dimensions, different limiting behavior, and different physical meaning. In particular, a maximum capacity should only be reported when the selected model contains a finite saturation plateau. Non-saturating models such as Freundlich may be useful empirical descriptions over a limited concentration range, but they do not define an intrinsic $Q_{\max}$ and should not be used to rank materials by maximum uptake.

To address this problem, we proposed a unified reporting framework that separates model fitting from descriptor extraction. The final result of an isotherm analysis should be reduced, whenever possible, to a common descriptor set:

$$\{Q_{\max}, K^*_{\text{aff}}, E_{\text{ads}}, \sigma_H\}.$$

Here, $Q_{\max}$ describes the finite adsorption capacity, K^*_{aff} places affinity on a dimensionless standard-state scale, E_{ads} expresses this affinity as an adsorption-energy descriptor, and σ_H quantifies the reported model-implied surface heterogeneity. This fingerprint allows adsorption data obtained from different isotherm models to be compared in a physically meaningful way, provided that the standard-state concentration C^0, experimental conditions, fitted parameters, and uncertainty metrics are explicitly reported.

The framework also provides a practical route for model selection. Rather than relying on R^2 alone, the proposed workflow combines non-linear regression, residual-based metrics, information criteria, nested-model tests, and physical validation based on limiting behavior and energy-distribution families. This approach helps distinguish cases where Langmuir is sufficient from cases where heterogeneous models such as Sips, Tóth, UNILAN, Jovanovic, or double Langmuir are genuinely justified. It also prevents over-interpretation of small statistical improvements that do not correspond to physically meaningful differences in adsorption behavior.

Finally, the descriptor-based structure of the framework makes adsorption data more suitable for cross-study comparison, uncertainty analysis, and machine-learning classification. By converting model-specific fitting results into a common set of thermodynamic and heterogeneity descriptors, adsorption studies can move from isolated parameter tables toward reproducible structure–performance relationships across adsorbents, contaminants, and water matrices. The proposed guidelines are therefore not intended to impose a single preferred isotherm model, but to ensure that whichever model is used, the reported quantities remain transparent, comparable, and physically interpretable.

Acknowledgement

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Supporting Information Available

The Supporting Information contains the full isotherm equations and limiting cases, residual error metrics, information criteria, model-selection workflow, energy-distribution derivations, descriptor-extraction tables, reporting protocol, and documentation of the companion notebooks.

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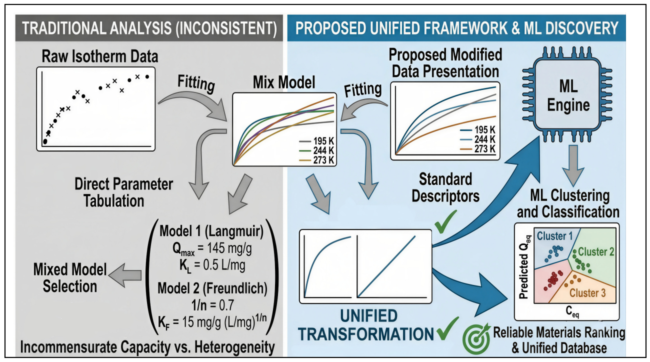
TOC Graphic


TRADITIONAL ANALYSIS (INCONSISTENT)
PROPOSED UNIFIED FRAMEWORK & ML DISCOVERY
Raw Isotherm Data
Fitting
Mix Model
Fitting
Proposed Modified Data Presentation
ML Engine
195 K
244 K
273 K
Direct Parameter Tabulation
Standard Descriptors
ML Clustering and Classification
Model 1 (Langmuir)
Q_{max} = 145 mg/g
K_L = 0.5 L/mg
Model 2 (Freundlich)
1/n = 0.7
K_F = 15 mg/g (L/mg)$^{1/n}$
Mixed Model Selection
Cluster 1
Cluster 2
Cluster 3
Predicted Q_{eq}
C_{eq}
UNIFIED TRANSFORMATION
Reliable Materials Ranking & Unified Database
Incommensurate Capacity vs. Heterogeneity